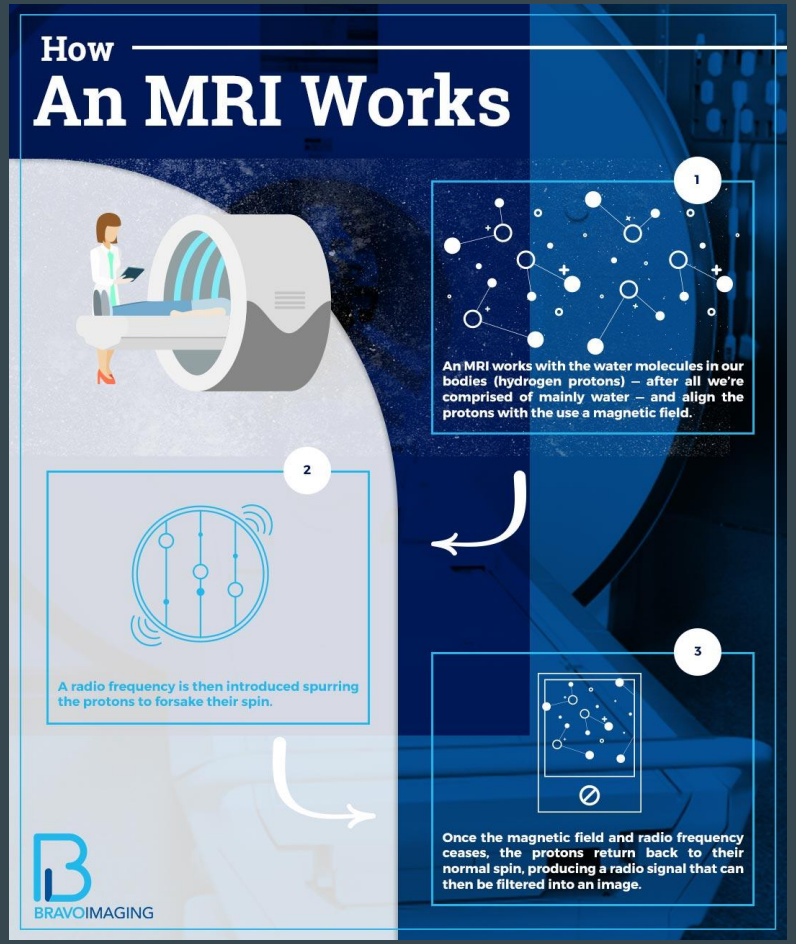
MRI Reconstruction

...

Background

MRI machines don't actually take pictures: they use magnets and the physical properties of water in our bodies to capture radio-frequency data.

Each radio signal is converted to a pixel in an image using a Fourier transform. By convention, we say that the MRI stores radio-frequency data in *k-space*, before being converted to an image.



Source: Bravo Imaging

The Problem

There are a lot of pixels in a typical MRI image, which means the machine has to send out a lot of radio-frequencies, which **takes a long time** . This is **expensive** for hospitals operating MRI machines and **uncomfortable** for patients, who have to stay still while images are taken.

Staying still for a long time is hard for

- Children
- Dementia Patients
- Patients with Huntington's, addictions, etc.

Shorter scan times = more accessible treatment

Potential Solution

What if, instead of measuring every value in k-space, we only directly measured a subset of them?

- Quicker scans, but more blurry images

Can we can **reconstruct** a partially formed MRI image using optimization?

- Quick scans with more detailed images

Mathematics

A = MF

- M = Diagonal Matrix (entries are 1 if frequency is being measured, 0 if not)
- F = Discrete Fourier Transform

Given observed undersampled k-space data *y_obs,* we are looking for α that solves

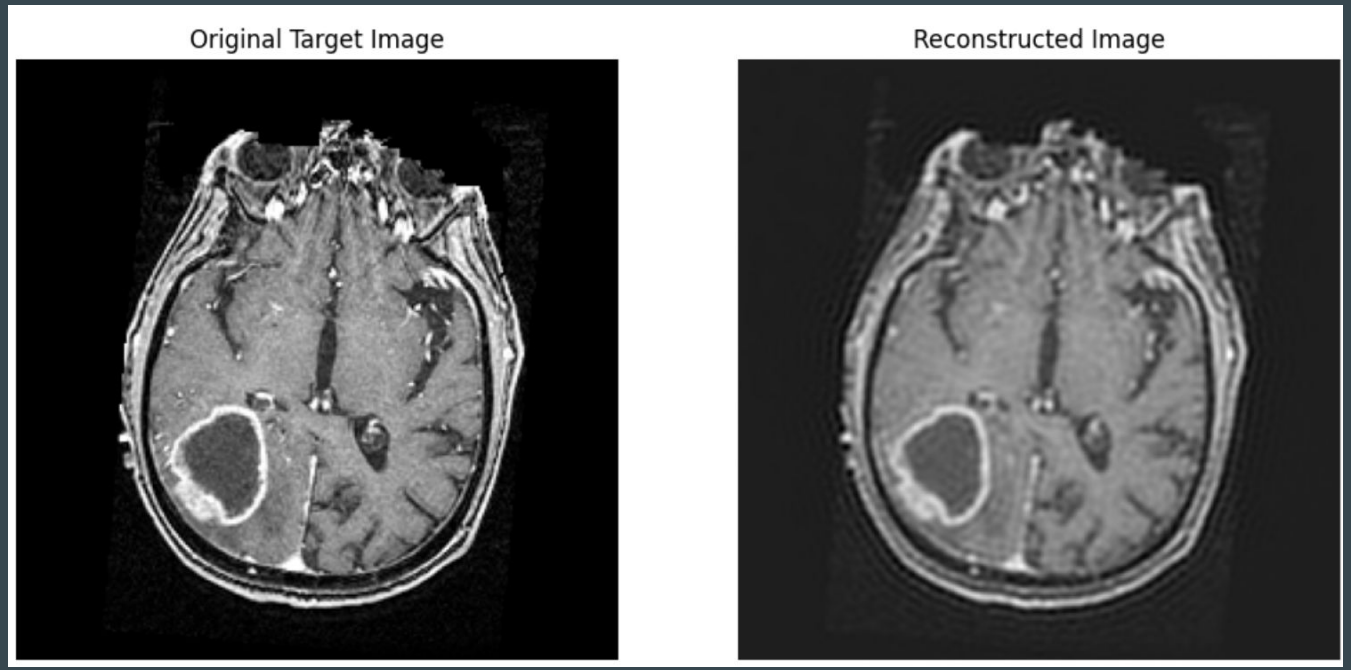
$$\min_{\alpha \in \mathbb{R}^n} ||A(K\alpha) - y_{obs}||_2^2$$

There are many possible solutions, so we restrict the reconstruction to a class of kernel basis functions for uniqueness. We use **Gaussian kernels** to impose smoothness.

Implementation

- **Data** : UPenn-GBM Cancer Imaging Archive. Selected target image.
  - Create kernel basis & lots of data preparation
- **Simulate** the under-sampling of frequencies: 'masking' the data by adding a filter to only show low-frequency waves.
- **Solve** : Using an Adam (Adaptive Moment Estimation) optimizer.
- **Validate** : (1) compare reconstructed image to original MRI target image and (2) measure loss over iterations.

Results



Pro:

- Working prototype w/real data

Con:

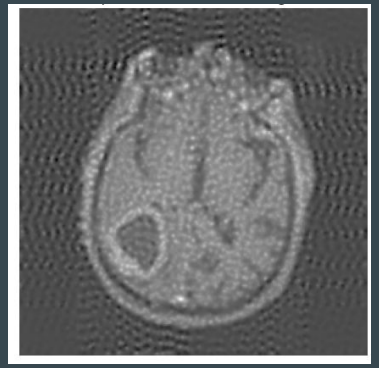
- Blurry image
- No frame of reference for accuracy

Comparing Kernels vs. Baseline Reconstruction

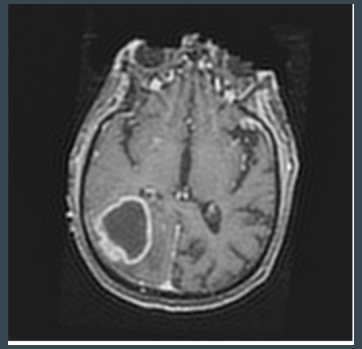
Metric	*LaPlacian Kernels*	*Gaussian Kernels*	*Zero-Filled FFT*
MSE	0.0186	0.00529	0.010049
PSNR	17.30 dB	22.78 dB	19.98 dB

Original Image

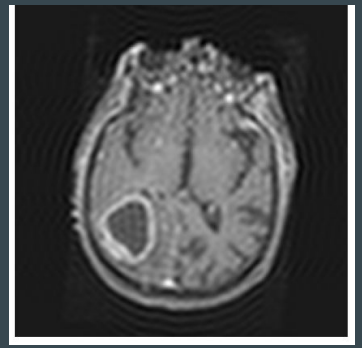
Laplace Reconstruction



Gaussian Reconstruction



Zero-filled FFT Reconstruction

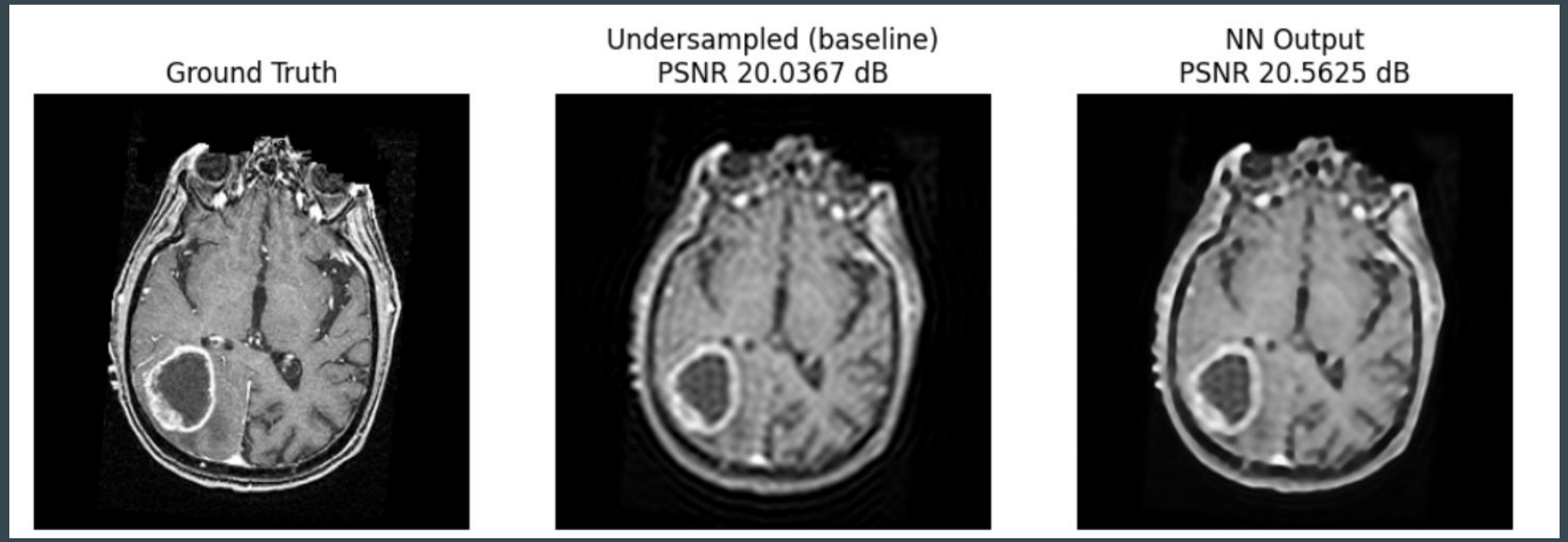


Gaussian kernel reconstruction provides highest clarity.

Another Baseline: The Traditional Approach

- Modern MRI reconstruction often uses learned priors (CNNs/diffusion models)
- Trained a small CNN:
  - 3 layers, 64 channels, 3x3 kernels (standard, small)
  - Input: undersampled MRI slice
  - Predicts a correction: $x* = x + f(x)$ where
    - x^* is final reconstructed image
    - $f(x)$ is learned correction
    - x is the undersampled input
  - Used L1 loss (common in image restoration tasks): $||x + f(x) - x_{true}||$

Neural Network Baseline Results



Gaussian kernel still achieves highest PSNR, implying the kernel method is competitive to traditional, learned methods.

Next steps

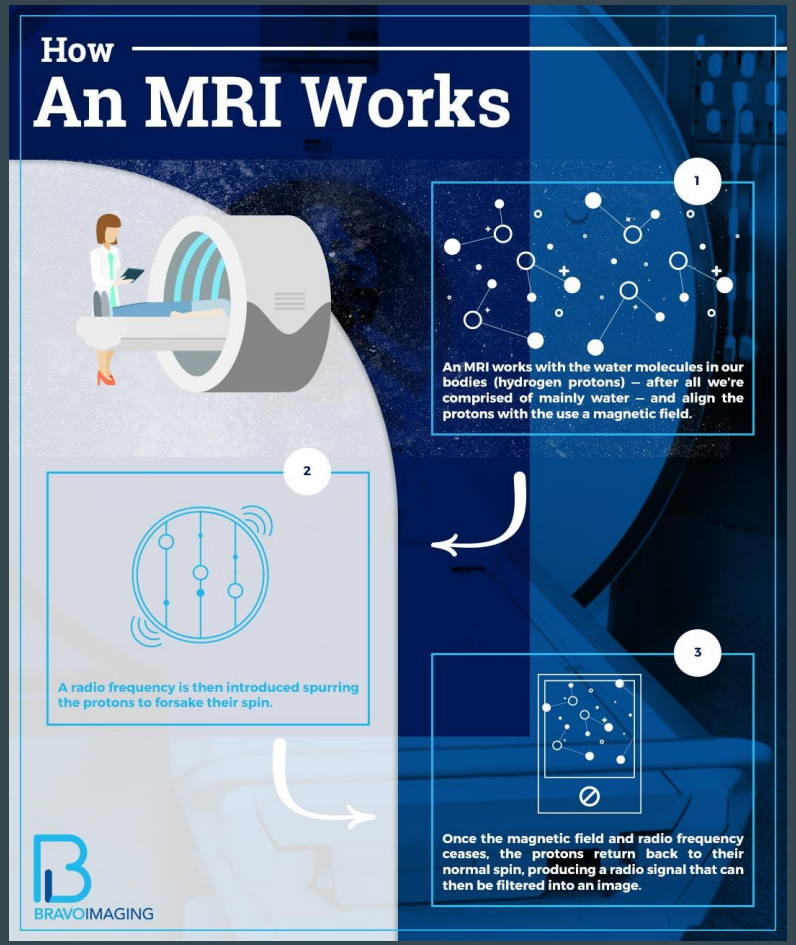
- **Improve Validation:** How good is our model vs. if we filled all empty pixels with blank space? We run a simulation with zero-filled FFT to simulate the null case.
  - Provides context for our reconstruction quality
- **Test with different kernels:** LaPlace kernel allows for sharper lines, which may create a more accurate image.
- **Combine with neural networks:** Neural network trained on reconstructed image to work as residual corrector.

Lightning Presentations: MRI Reconstruction

Problem Statement & Motivation

There are a lot of pixels in a typical MRI image, which means the machine has to send out a lot of radio-frequencies, which **takes a long time** . This is **expensive** for hospitals operating MRI machines and **uncomfortable** for patients, who have to stay still while images are taken.

Shorter scan times = more accessible treatment



Technical Approach

A = MF

- M = Diagonal Matrix (entries are 1 if frequency is being measured, 0 if not)
- F = Discrete Fourier Transform

Given observed undersampled k-space data *y_obs,* we are looking for α that solves

$$\min_{\alpha \in \mathbb{R}^n} ||A(K\alpha) - y_{obs}||_2^2$$

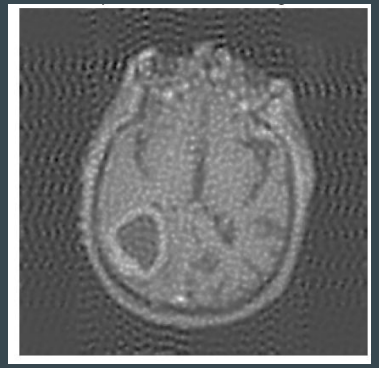
There are many possible solutions, so we restrict the reconstruction to a class of kernel basis functions for uniqueness. We start with **Gaussian kernels** to impose smoothness.

Initial Results pt. 1

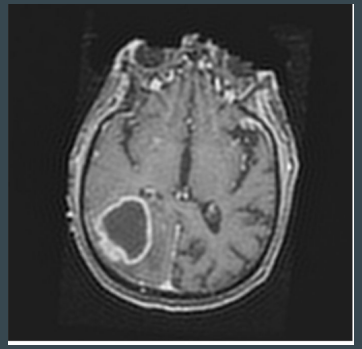
Metric	*LaPlacian Kernels*	*Gaussian Kernels*	*Zero-Filled FFT*
MSE	0.0186	0.00529	0.010049
PSNR	17.30 dB	22.78 dB	19.98 dB

Original Image

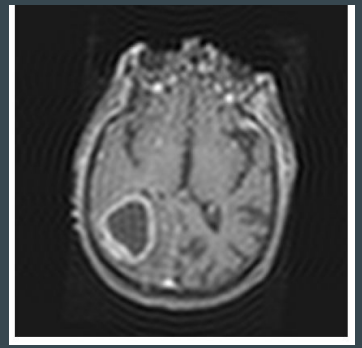
Laplace Reconstruction



Gaussian Reconstruction



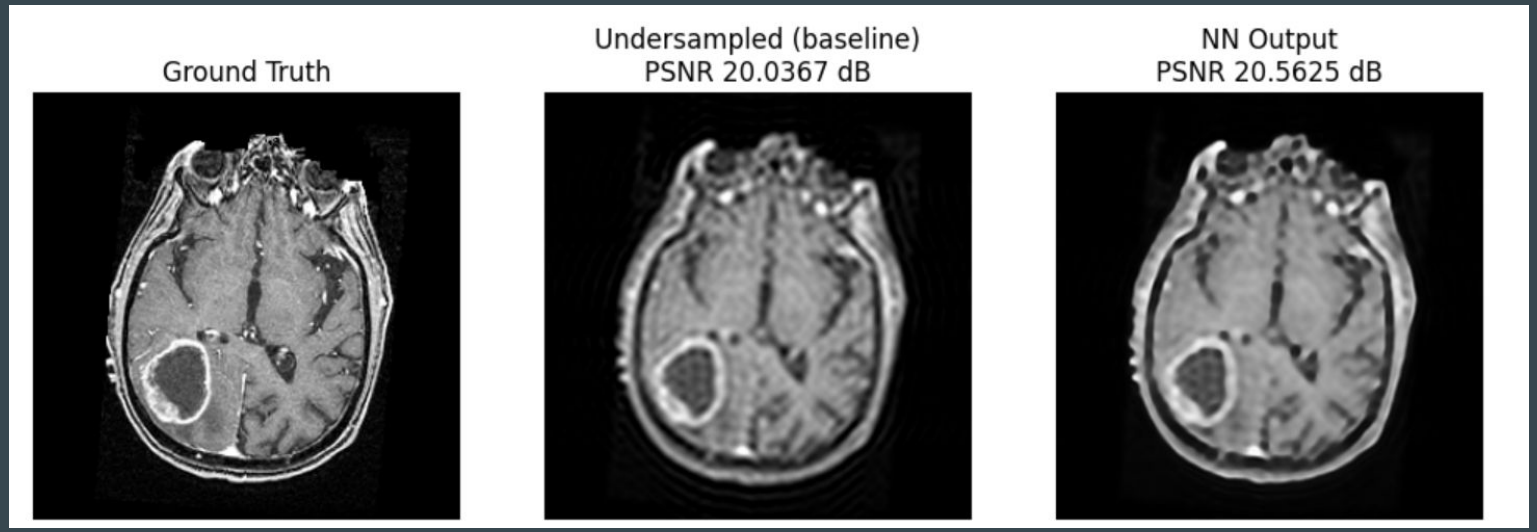
Zero-filled FFT Reconstruction



Gaussian kernel reconstruction provides highest clarity.

Initial Results pt. 2

- Modern MRI reconstruction often uses learned priors (CNNs/diffusion models)
- Trained a small residual CNN, then reconstructed the same slice we used the kernel method on



Gaussian kernel still achieves highest PSNR, implying the kernel method is competitive to traditional, learned methods.

Next Steps

- Moving toward more general image reconstruction/upscaling due to difficulty in determining medical benchmarks
- Combining the kernel approach with neural networks
  - Use kernels to get a smooth reconstruction
  - Feed the reconstruction into a neural network
  - Network learns the residual and works as a residual corrector
  - Can only modify the unobserved regions of k-space
- Using the Structural Similarity Index as another metric of success

$$SSIM(I_{LR}, I_{HR}) = \frac{(2\mu_{I_{LR}}\mu_{I_{HR}} + c_1)(2\sigma_{I_{LR},I_{HR}} + c_2)}{(\mu_{I_{LR}}^2 + \mu_{I_{HR}}^2 + c_1)(\sigma_{I_{LR}}^2 + \sigma_{I_{HR}}^2 + c_2)}$$

- Explore different deep learning and neural network upsampling strategies (Pre, Post, and Progressive)